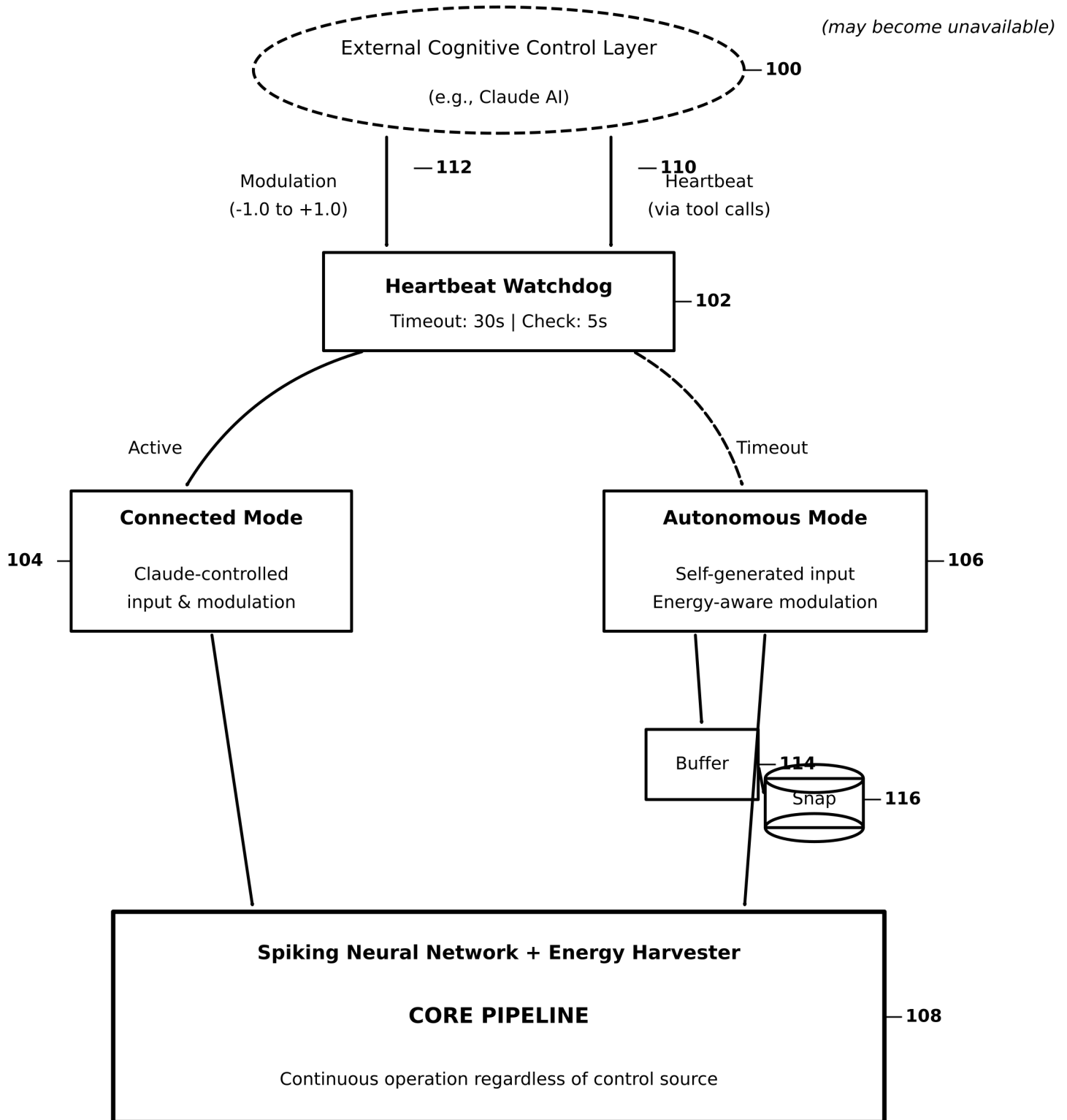


System Architecture with Cognitive Fallback

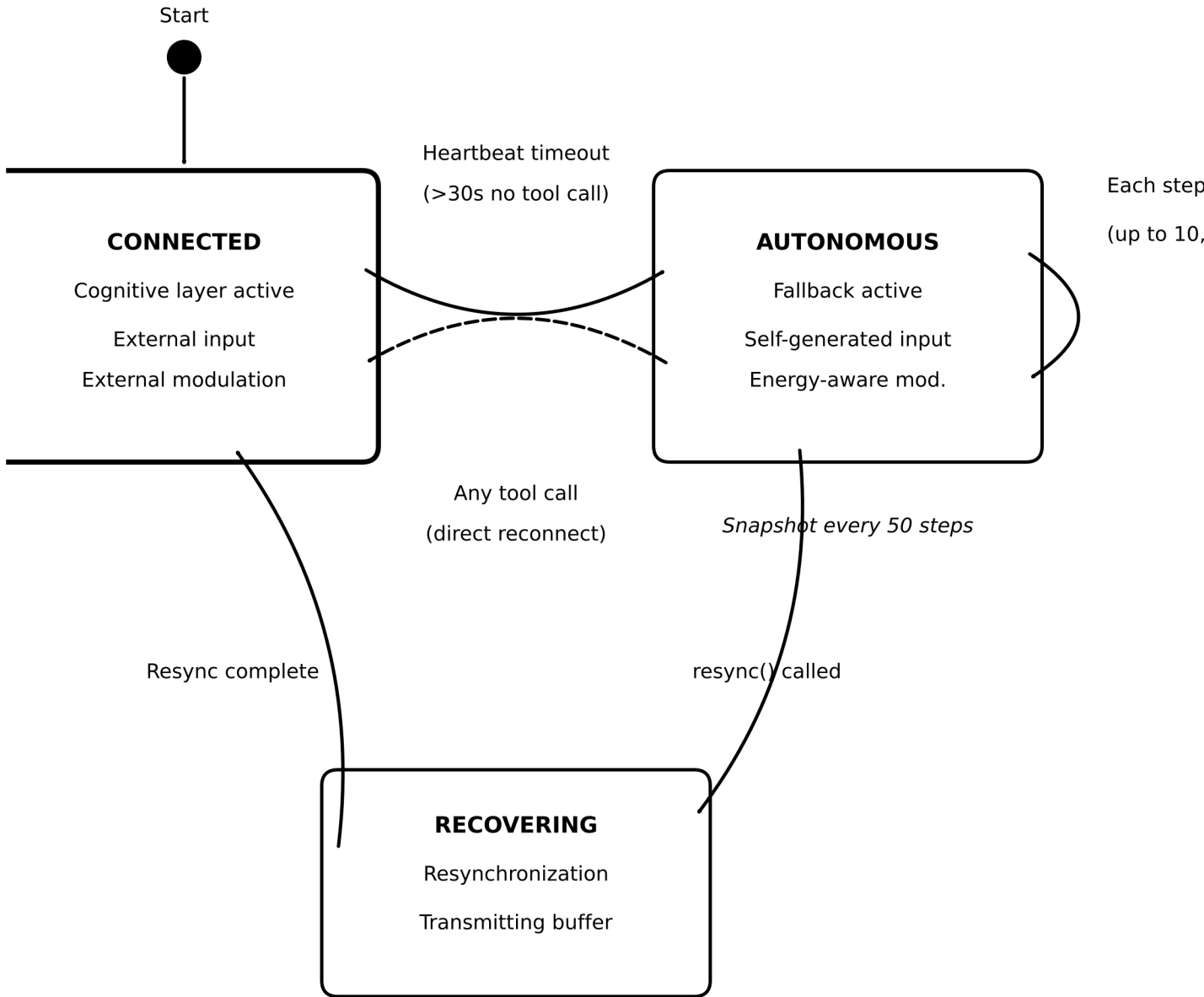


THE SYSTEM NEVER STOPS

FIG. 1

Control source changes; computation continues seamlessly

State Transition Diagram

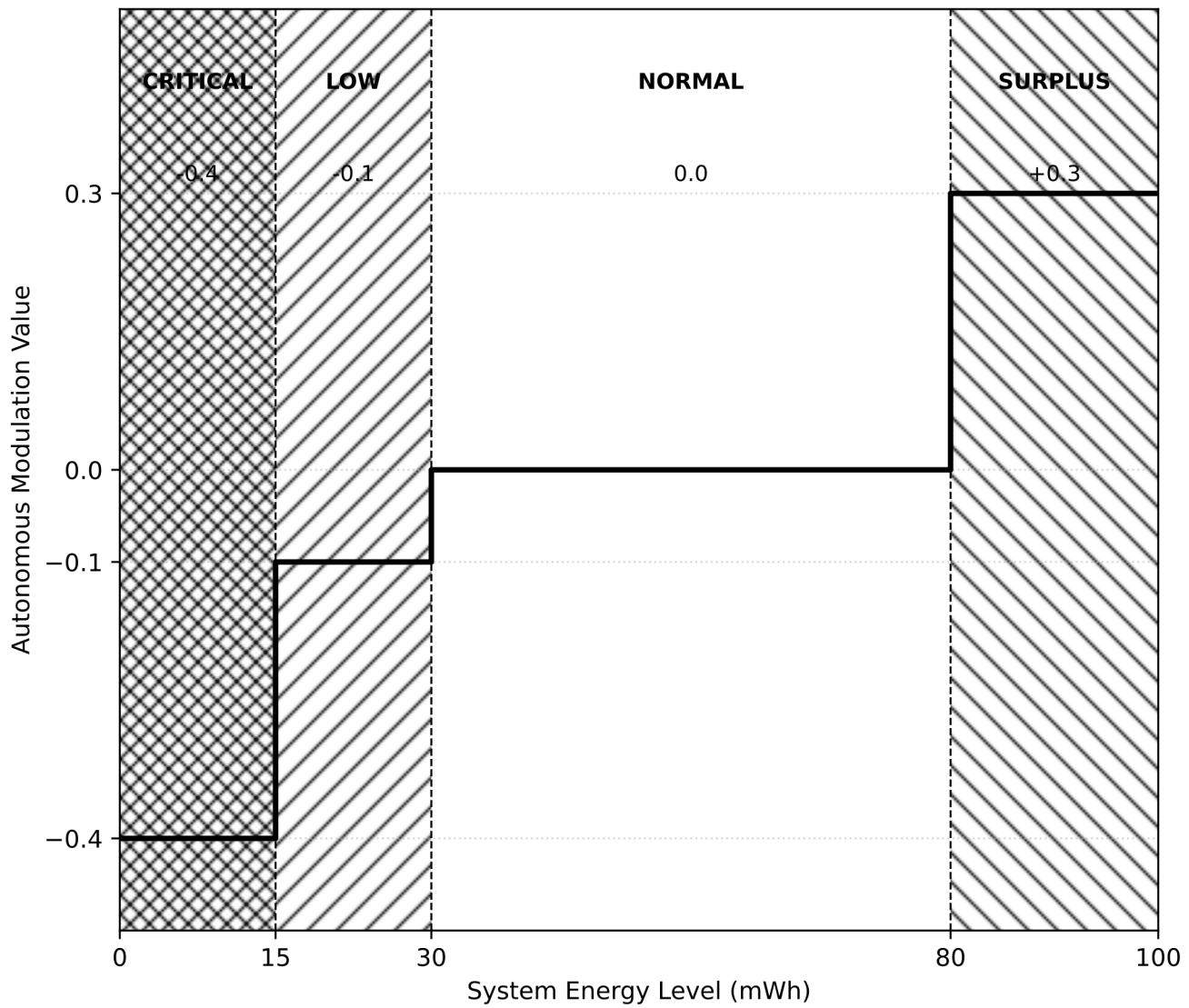


Hard cap: 10,000 autonomous steps

———— Standard transition - - - - - Direct reconnection

FIG. 2

Energy-Aware Modulation Curve (120)



The system autonomously adjusts processing intensity based on energy

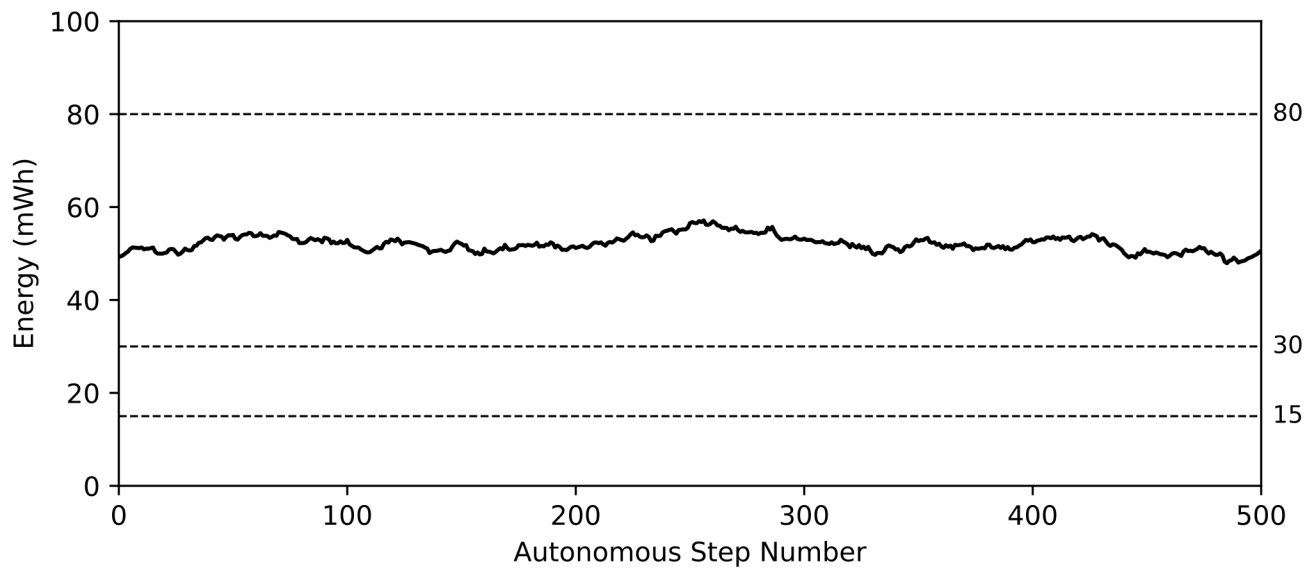
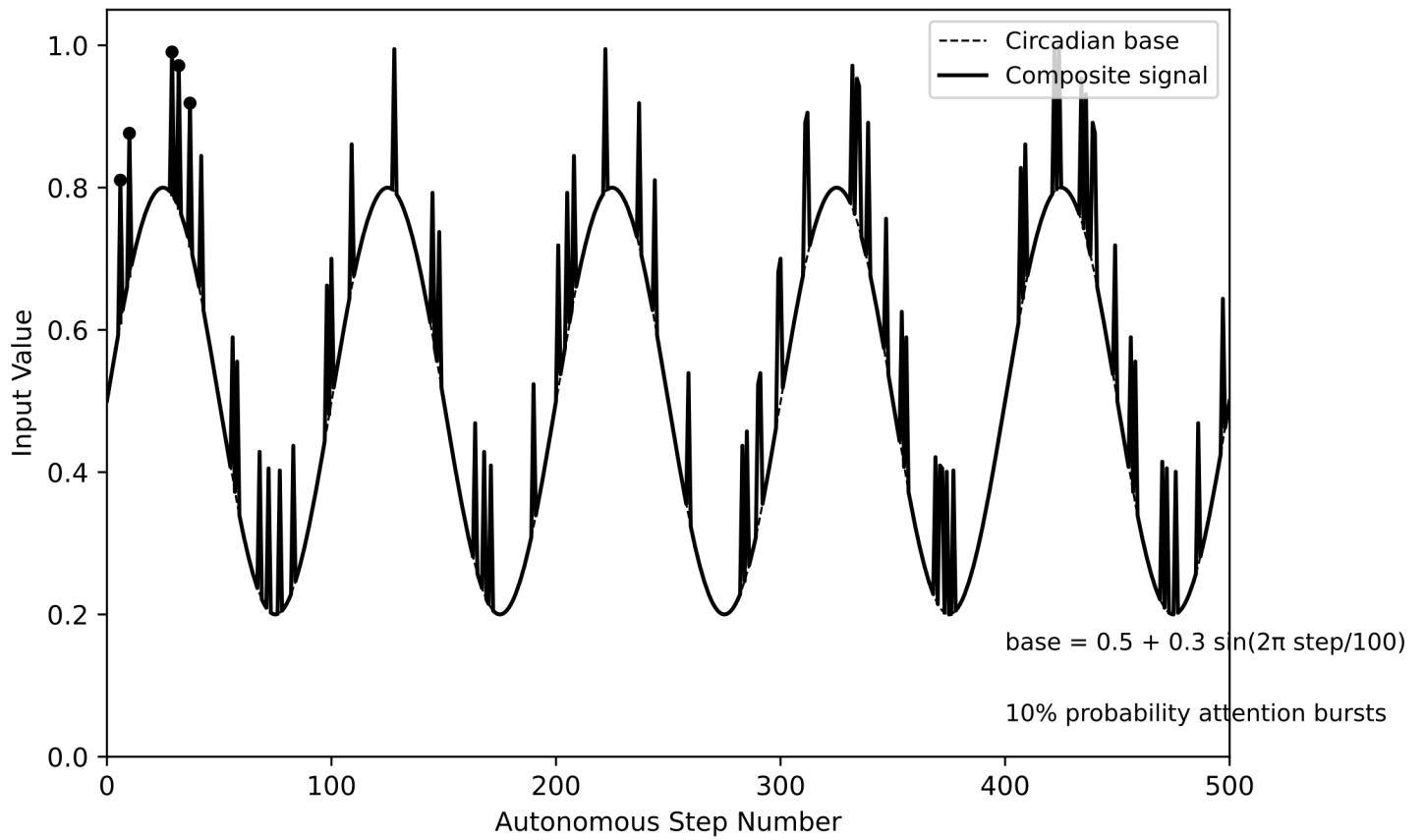
CRITICAL: Maximum conservation to prevent shutdown

SURPLUS: Opportunistic exploration using excess energy

Mimics biological metabolic regulation

FIG. 3

Autonomous Input Generator Output (118)



Circadian rhythm provides structured temporal variation

Energy level determines modulation via curve in FIG. 3

FIG. 4

Resynchronization Payload Structure (122)

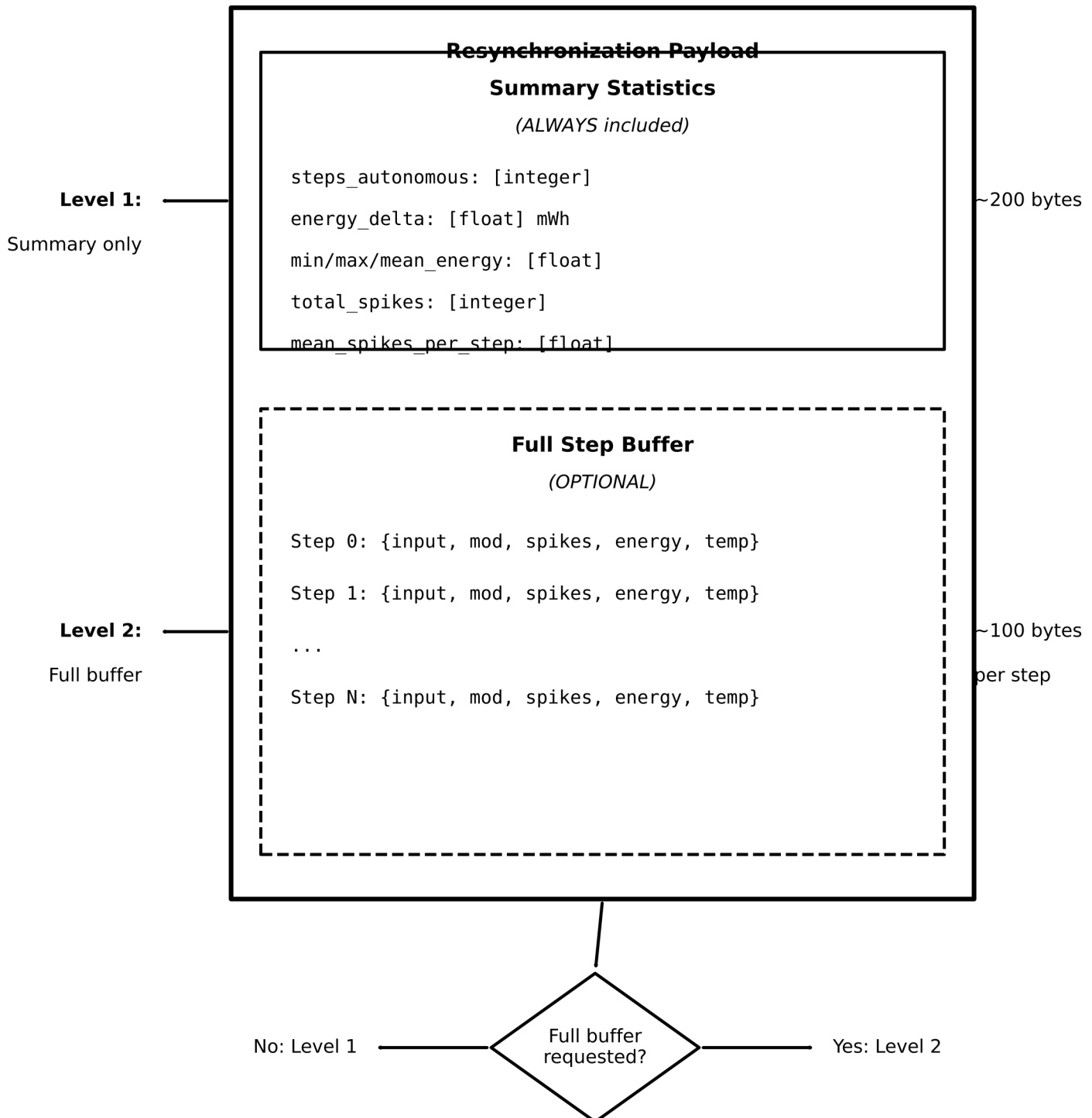
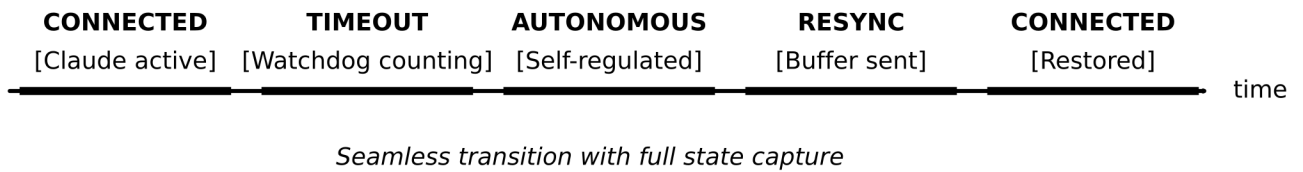


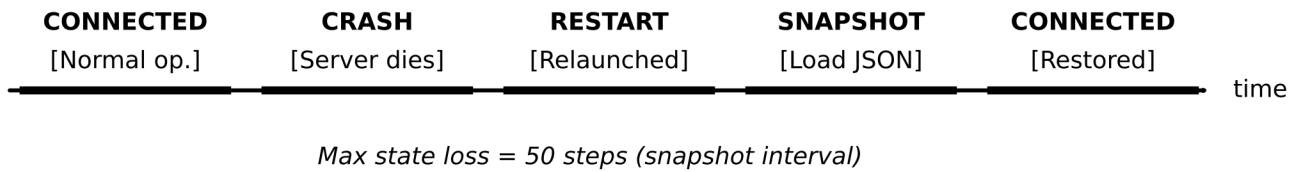
FIG. 5

Recovery Timeline Diagrams

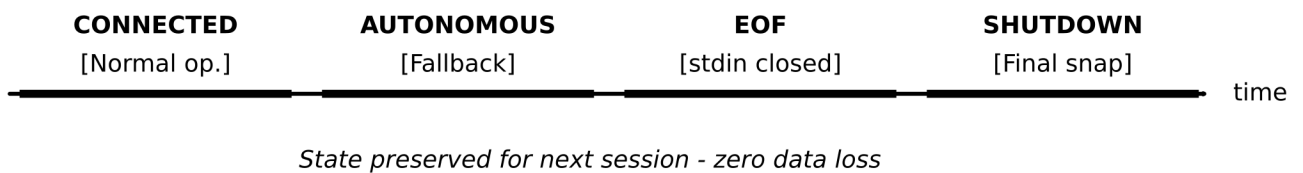
Timeline A: Normal Reconnection



Timeline B: Crash Recovery



Timeline C: Clean Shutdown



Legend:

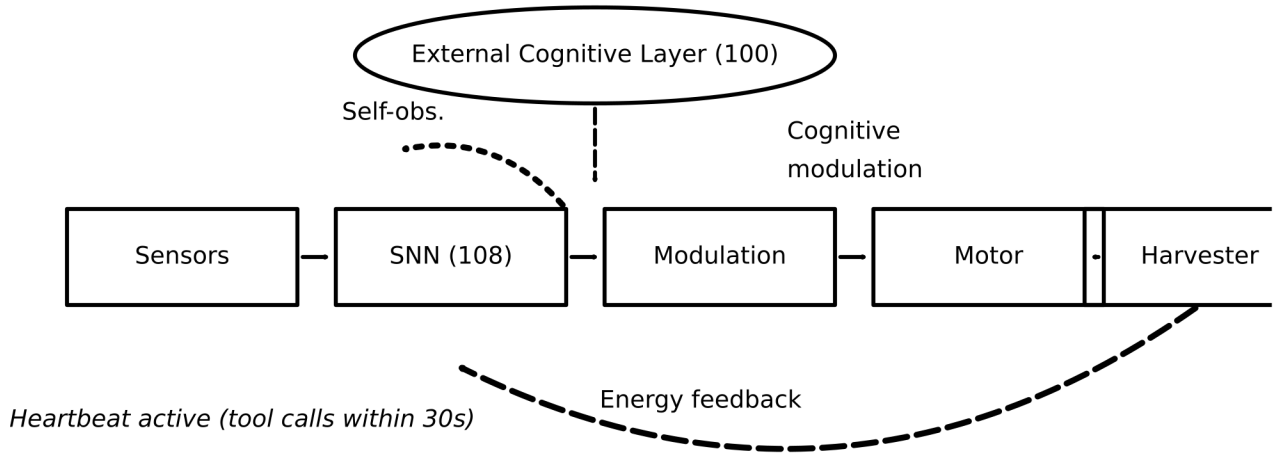
———— System actively processing

Bold markers indicate NO DATA LOSS at transitions

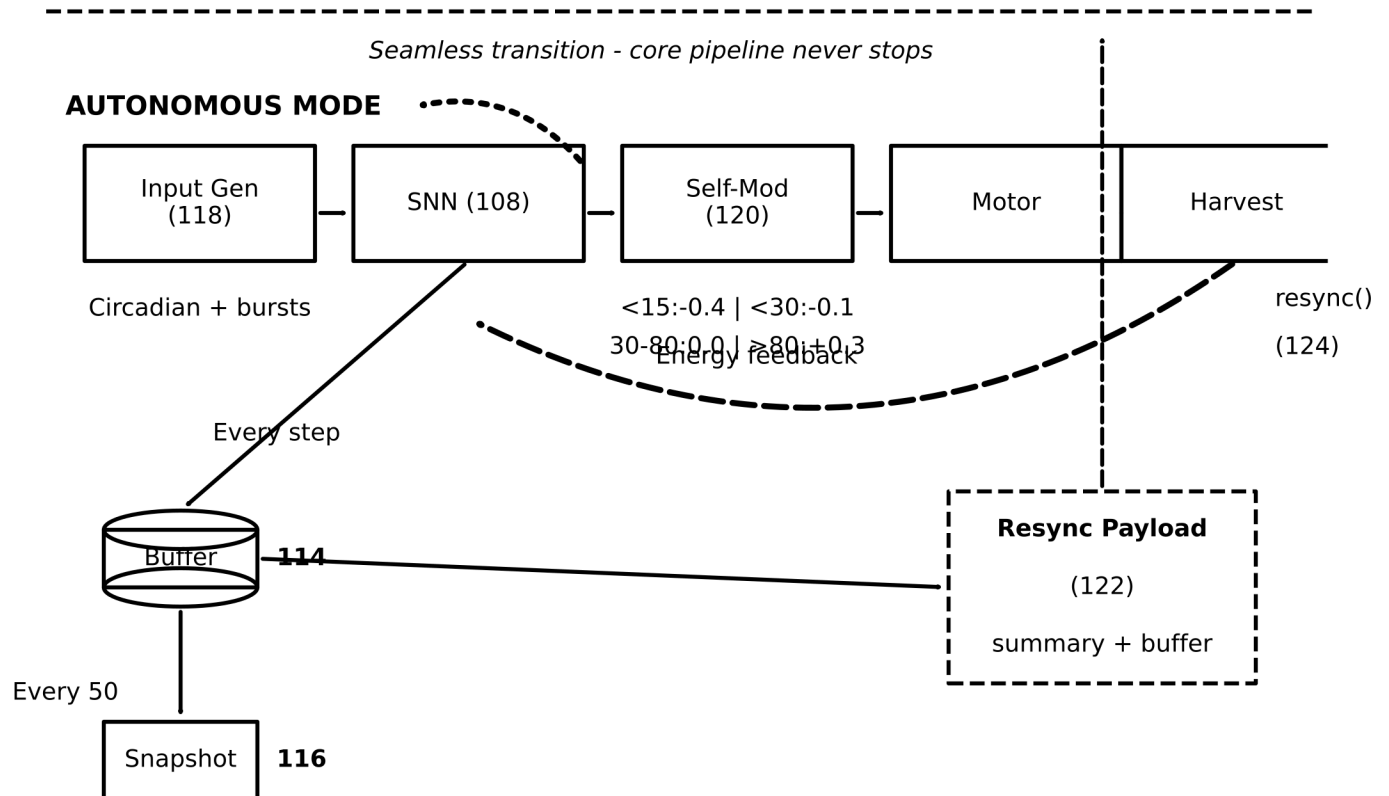
FIG. 6

End-to-End Signal Flow

CONNECTED MODE



DISCONNECTION EVENT (heartbeat timeout > 30s)



The fallback changes INPUT SOURCE and MODULATION SOURCE

Core neural processing, motor actuation, and energy harvesting

continue WITHOUT INTERRUPTION through the transition

FIG. 7

Forward

Energy

.....

Self-obs.